commit;

(operations) begin → Transaction → commit (Transaction succeeded)

initial state

start transaction; → Rollback (Transaction failed)

rollback;

Transactions
↳ stock trading, ATM Withdrawals, bank transfer, e-commerce orders, online payment

Alina
↳ 1000
(credit) +50
(1000) 1050
Still pending

Alex
↳ 500
−50 (deducted)
450

↳ withdrawal
↳ Deposit → succeed / fail

transfer

(Source → destination)
(Axis)

commit
rollback

Transaction
verify.

Hdfc bank
20,000
−1000
19000

(Source)

if failed
↳ Savepoint.

Savepoint

series of transactions, namely of steps

1
2
3
4
5
6

Savepoint.
↳ checkpoint.

checkpoint 1
checkpoint 2
checkpoint 3
checkpoint 4
checkpoint 5
Source

ER - Entity Relationship Diagram

↳ Entity (Table) → Employee

Strong primary key

Weak primary key

Attribute
- non-primary key
- id (primary key)
- age (derived attribute)

↳ Relationship

Employee — works for — company

ER Diagram
↳ Doctor Appointment App
↳ Graduate App
↳ Edtech

Customer — buying — Product

ACID Properties

A → Atomicity
↳ a transaction fail or succeed.

C → Consistency
↳ Data Validity (constraints, pre-defined rules)

I → Isolation
↳ if multiple transaction is happening, it should not impact each other

D → Durability
↳ Persistant
if a transaction is committed, can't be deleted.

miro